Institute of Professional Studies

Software Engineer
INTERNSHIP SYNOPSIS

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GURU GOBIND SINGH INDRAPRASTHA UNIVERSITY

Academic Year: 2025–2026

Introduction

- Internship completed under MetaCyrus, a technology-driven startup focused on product development and digital solutions.

- Worked primarily onVenu(venue booking platform) and HealthLine (HealthCare Product platform).
- Gained practical exposure to real-world software development and web application workflows.
- Contributed in areas such as frontend development, UI/UX enhancement, API integration, authentication, and performance optimization.
- Improved understanding of modern frontend technologies including React.js, Next.js, TypeScript, and Tailwind CSS.
- Learned how professional teams manage projects through collaboration, daily sync-ups, version control, and structured workflows.
- Applied academic knowledge in a practical environment and developed strong technical and problem-solving skills.

Problem Statement

- Building the Venu frontend from scratch required careful architecture planning, component design, and feature prioritization.
- Implementing secure authentication (Google OAuth and email-based) and protected routing was a significant technical challenge.
- Managing role-based access control for Durian with separate vendor and admin dashboards required conditional rendering and permission management.
- Maintaining consistent and responsive UI/UX design across different devices required continuous improvement and refinement.
- Debugging API integration issues, handling asynchronous operations, and identifying root causes were major challenges during development.
- Ensuring application performance, reducing API calls, and improving web vitals was an important concern.
- Adapting to modern tools, frameworks, and professional development practices within limited time was challenging.

Objectives

- To gain practical exposure to real-world software development
- To apply academic knowledge to Venu and Durian platforms
- To improve frontend skills: ReTo gain practical exposure to real-world software development and web application workflows.
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Methodology

- Analyzed the project requirements and product workflows of Venu and Durian before implementation.
- Followed a systematic approach involving requirement understanding, development, testing, and optimization.
- Used a modular and component-based development approach for better scalability and maintainability.
- Worked on UI/UX improvements using Figma design planning and responsive development techniques with Tailwind CSS.
- Implemented and integrated functionalities through frontend-backend communication using REST APIs and Axios.
- Used Git and GitHub for version control, feature branching, and collaborative development.
- Conducted research and self-learning through assigned courses (Advanced Next.js, Django Basics).
- Focused on improving application performance, responsiveness, and user experience through React Query caching.

Tools and Technologies

Software
- Visual Studio Code was used as the primary code editor for writing, debugging, and maintaining application code.
- GitHub was used as a collaborative platform for code hosting, and repository management.

Languages
- JavaScript and TypeScript were used as the primary programming languages for frontend development.
- HTML for structuring web pages, CSS and Tailwind CSS were used for styling and ensuring responsive interfaces.

Platforms & Frameworks
- React.js and Next.js were used for building dynamic frontend user interfaces and reusable components.
- React Query was used for server state management, caching, and data synchronization.
- Axios was used for API handling and frontend-backend communication.
- Google OAuth 2.0 and JWT were used for secure authentication.

Learnings

- Gained practical understanding of real-world software development workflows and project execution in a startup environment.
- Improved technical knowledge in frontend development, UI/UX design, API integration, authentication, and performance optimization.
- Learned how modern web applications are developed using React.js, Next.js, TypeScript, and Tailwind CSS.
- Developed problem-solving and analytical skills while handling real-time development challenges like dynamic guest count and conditional profile states.
- Understood the importance of writing clean, scalable, and maintainable code using reusable components and utility functions.
- Learned effective debugging and testing techniques to improve application performance and reliability.
- Improved teamwork, communication, and collaboration skills through daily sync-ups, Slack communication, and interaction with pod leads.

Conclusion

- The internship at WhatBytes provided valuable exposure to real-world software development practices and product engineering.
- Working on the Venu and Durian platforms helped in understanding modern frontend development in a professional environment.
- Enhanced technical skills in frontend development, API integration, authentication, performance optimization, and responsive design.
- Improved problem-solving abilities, analytical thinking, and adaptability while handling real-time challenges.
- Gained experience in collaborative development using Git, daily sync-ups, professional workflows, and role-based access control.
- Learned the importance of clean coding, reusable components, performance optimization (30% web vitals improvement), and user-centric application design.
- Achieved measurable improvements including 38% reduction in API calls using React Query.
- The internship successfully bridged the gap between theoretical knowledge and practical implementation.

THANK YOU